Plumage: Fine-Grained Bird Species Classification on CUB-200-2011

CSE485/488 Deep Learning – Milestone 1 Report

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November 2026

Abstract

In this milestone we worked on the CUB-200-2011 dataset, which has 11,788 bird images from 200 species. We first did an exploratory data analysis and used what we found to decide on the splits, the cleaning and the augmentation. Then we trained a CNN from scratch as a baseline and compared it with a ResNet-50 pretrained on ImageNet, once with a frozen backbone and once fully fine-tuned. The scratch CNN reached 40.2% test accuracy, the frozen ResNet-50 64.3% and the fine-tuned ResNet-50 82.9%, which is close to what published papers report for ResNet-50 at the same input size. We also tried grid and random search for the learning rate and weight decay, but it did not improve on our default setting. Finally, we used Grad-CAM to see where each model looks and found that the scratch CNN often relies on the background while the ResNet-50 focuses on the bird itself.

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1 Introduction

The goal of the whole project is to build one visual encoder and reuse it for three tasks: species classification (this milestone), and later image colorization and image captioning (Milestone 2). For this reason we tried to write the code so that the encoder can be swapped with one line in a config file, and so that it exposes its feature map for later use.

We chose the CUB-200-2011 dataset [1] with the captions from Reed et al. [2]. Every image has a species label and 10 human-written captions, so the same dataset works for all three tasks. It is also small enough (about 1.2 GB) to train on the free T4 GPUs on Kaggle and Colab.

The main difficulty is that the task is *fine-grained*. Many species look almost the same and only differ in small details like wing bars, the eye ring or the shape of the bill, and there are only around 30 images per species. So we expected a model trained from scratch to struggle, and that is one of the things we wanted to show.

The work was split into five steps, one per team member, as shown in Table 1. Each person worked on their own branch, opened a pull request, and the previous person reviewed it.

Member	ID	Step	Main work
Jana Hany Elshafie	22P0235	1	Repository structure, YAML configs, dataset class, validation split, MLflow wrapper, CLI
Habeba Adel Sallam	22P0259	2	EDA, duplicate detection, caption cleaning, preprocessing and augmentation
Nouran Mokhtar Elsayed	22P0254	3	Scratch CNN, training loop, evaluation metrics, training curves, baseline run
Mohamed Yehia Zakaria	22P0064	4	ResNet-50 (frozen and fine-tuned), grid and random search, final model
Abdallah Belal Momen	22P0036	5	Grad-CAM, `predict` and `analyze` commands, error analysis, benchmark

Table 1: Division of work between team members.

2 Dataset and Exploratory Analysis

We load the dataset from Hugging Face (`Multimodal-Fatima/CUB_train` and `CUB_test`). All the numbers and figures in this section are produced by one command, `python -m src.cli eda`. Figure 1 shows some example images.

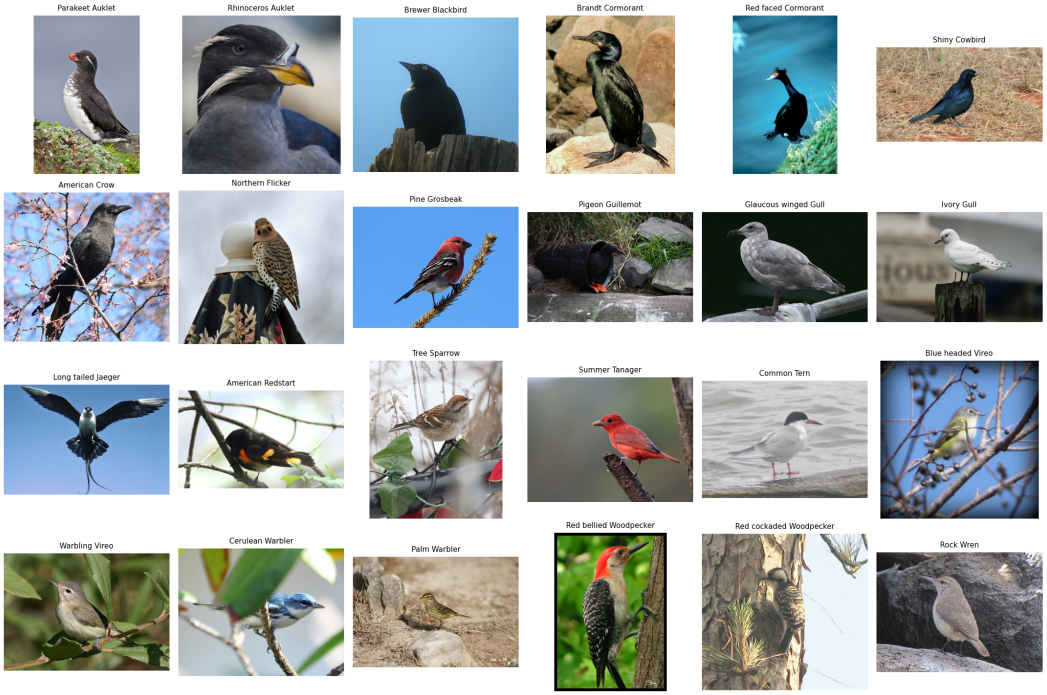


Figure 1: Random sample of training images with their species names.

2.1 Data splits

The dataset only comes with a train set (5,994 images) and a test set (5,794 images), so there is no validation set. If we chose our hyperparameters by looking at the test accuracy, the test number would be too optimistic. So we took 3 images per class (600 in total) out of the train set and used them as validation. We did the split per class, because with only ∼30 images per species a normal random split could leave some species with no validation images at all. The indices are saved in `configs/splits/val_indices.json` so that all of us used exactly the same split. The final split is in Table 2.

Split	Images	Used for
Train	5,389	training (5,994 − 600 val − 5 duplicates)
Validation	600	choosing models and hyperparameters
Test	5,794	final evaluation only

Table 2: Data splits used in all experiments.

2.2 Findings and the decisions we made

Class balance. The train set is almost perfectly balanced (26–27 images per class, imbalance ratio 1.04), but the test set is not (11–30 images per class, ratio 2.73), as can be seen in Figure 2. Because of this we did not use class weights, but we report **macro F1** next to accuracy, since macro F1 gives every species the same weight.

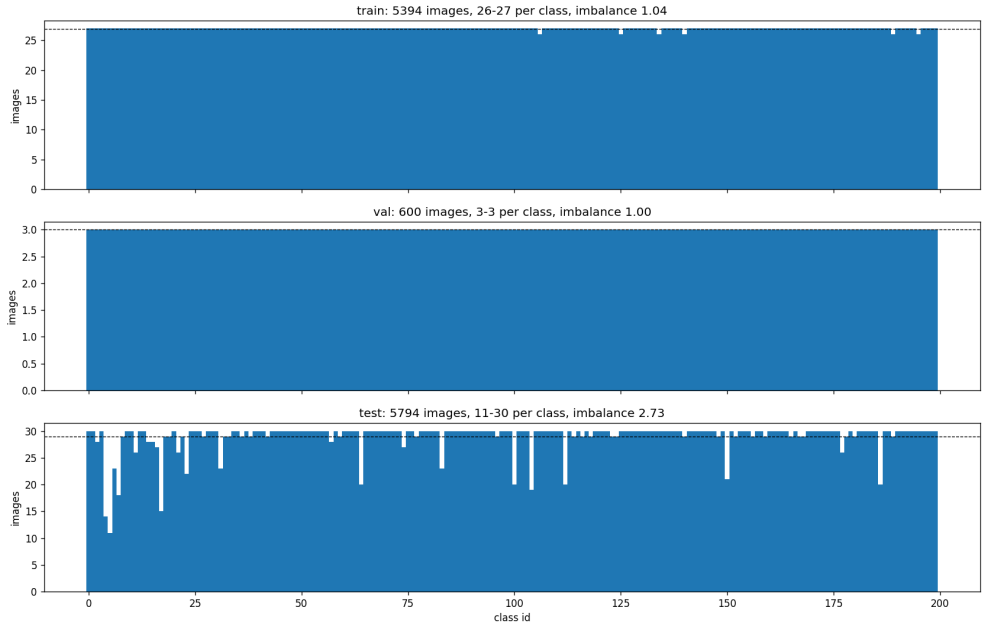


Figure 2: Number of images per class in the train, validation and test splits.

Image size and color mode. Almost all images have a longest side of 500 px (median 500×375), with aspect ratios between 0.38 and 3.6 (Figure 3). Only 148 images have a side smaller than 224 px, so resizing to 224 does not lose much. We found 8 grayscale images and no corrupted files, so we simply convert everything to RGB.

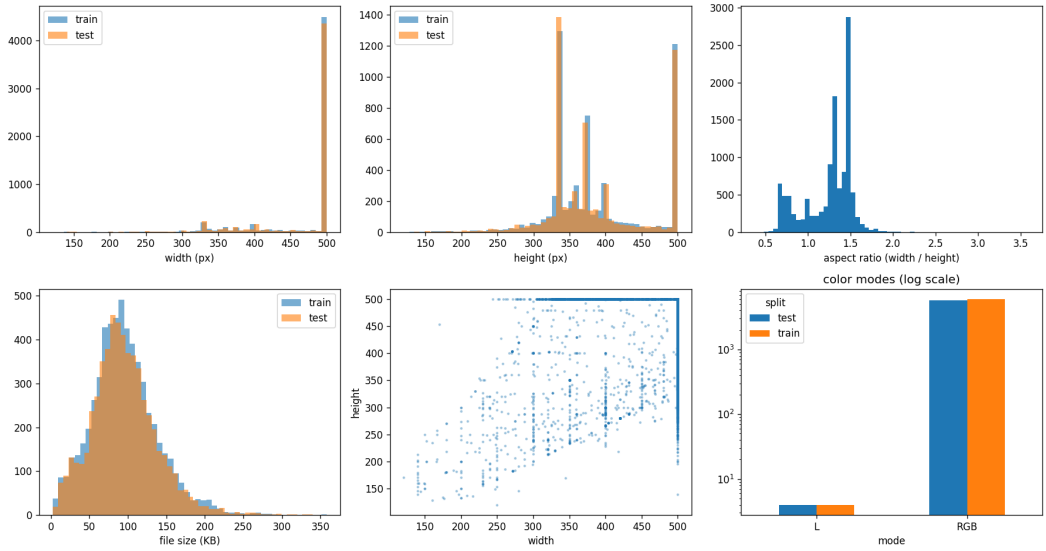
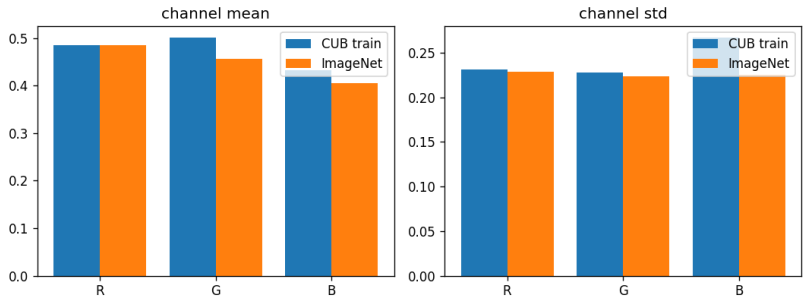


Figure 3: Image width, height, aspect ratio and file size distributions.

Duplicates. We used perceptual hashing to look for duplicate images. pHash (distance ≤ 4) gave 8 candidate pairs, but when we looked at them, 3 of them were actually different birds on a plain sky or snow background. So we added a second check with dHash: the real duplicates differed by 0–1 bits and the false matches by 15–17 bits. In the end we had 3 train–train pairs and **2 train–test pairs** (Figure 4a). The train–test duplicates are a problem because the model would see test images during training, so we removed 5 images from the train set.

Channel statistics. The mean of our train images is $(0.486, 0.501, 0.433)$ compared to ImageNet's $(0.485, 0.456, 0.406)$, so CUB is a bit greener and bluer, which makes sense with all the trees, sky and water (Figure 4b). We use the ImageNet values for pretrained models because they were trained with them, and the CUB values for the scratch CNN.

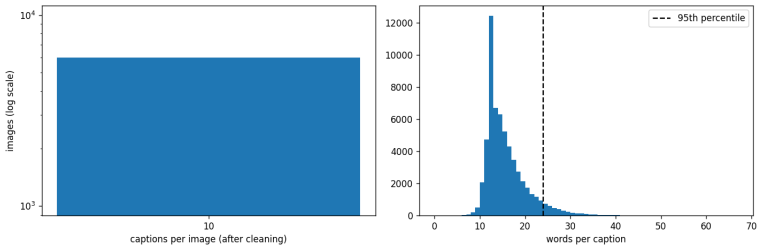
(a) Duplicate pairs



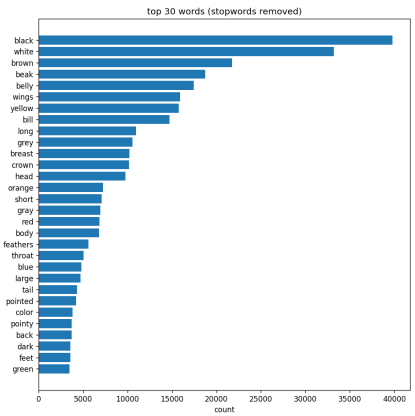
(b) Per-channel mean and std, CUB vs ImageNet

Figure 4: Duplicate images found with pHash + dHash (left) and channel statistics (right).

Captions. Even though captions are not used until Milestone 2, we analysed them now. Every image has exactly 10 captions, with an average length of 15.3 words and 95% of them under 24 words (Figure 5). About 100 captions had a broken Unicode replacement character in place of every space (e.g. "this??bird??has"), and there were also many `/`, `;` and `&` symbols. We wrote `clean_caption()` to fix these. After cleaning, the vocabulary went from 6,690 to 4,076 words, and only 1,243 words appear at least 5 times. Most of the rare words are typos (e.g. "abdomin", "ablack"), so in Milestone 2 we plan to keep words with frequency ≥ 5 and map the rest to `<unk>`.



(a) Caption length and captions per image



(b) Most frequent words

Figure 5: Caption statistics.

2.3 Preprocessing and augmentation

For training we use `RandomResizedCrop(224)` keeping 50–100% of the image, horizontal flips, rotation up to ±10°, and mild color jitter (brightness, contrast and saturation 0.2). We kept the hue jitter very small (0.02) because color is often what separates two species, for example the Summer Tanager and the Scarlet Tanager. We also did not use vertical flips since birds are not upside down in real photos. For validation and test we resize to 256 and take a center crop of 224. Figure 6 shows some examples.

Figure 6: One training image with 8 random augmentations.

3 Models

All encoders share the same interface: `forward_features()` returns the spatial feature map, `forward()` returns the pooled vector, and `get_gradcam_layer()` returns the last convolutional layer. The classifier is just the encoder followed by dropout (0.3) and one linear layer with 200 outputs. This made it easy to switch between models without changing the training code.

3.1 CNN from scratch

Our baseline is a simple VGG-style network with 5 blocks, each made of two Conv3×3–BatchNorm–ReLU layers followed by a 2×2 max pool (Table 3). At the end we use global average pooling instead of flattening, so the last layer is 512 → 200 instead of 25,088 → 200, which gives far fewer parameters and less overfitting. The total is 4.82M parameters.

Block	Channels	Output size
1	$3 \to 32$	112×112
2	$32 \to 64$	56×56
3	$64 \to 128$	28×28
4	$128 \to 256$	14×14
5	$256 \to 512$	7×7
GAP + dropout + linear	$512 \to 200$	1×1

Table 3: Architecture of the scratch CNN for a 224×224 input.

3.2 Pretrained ResNet-50

For transfer learning we used ResNet-50 [3] with the torchvision `IMAGENET1K_V2` weights, after removing its pooling and 1000-class layer. It gives a $2048 \times 7 \times 7$ feature map. We tried it in two ways:

- **Frozen:** the backbone is locked and only the last linear layer is trained (0.41M trainable parameters out of 23.9M). We also had to keep the backbone in eval mode, because otherwise BatchNorm keeps updating its running statistics even when the weights are frozen.
- **Fine-tuned:** all layers are trained, with a small learning rate (10^{-4}) so the ImageNet features are adjusted rather than destroyed.

4 Training Setup

All models were trained with the same function, `train_model()`, using cross-entropy loss, AdamW [5], a cosine learning rate schedule [6], batch size 32 and mixed precision on the GPU. We use early stopping on *validation accuracy* and keep the checkpoint of the best epoch. We chose accuracy instead of loss because with so few images per class the validation loss could keep going down even after the accuracy stopped improving. All runs were done on a Kaggle T4 GPU, since one epoch on a laptop CPU took about 16 minutes. Every run is logged to MLflow with its config, per-epoch metrics, curves and checkpoint.

Model	lr	Max epochs	Patience	Normalization
Scratch CNN	3×10^{-4}	40	10	CUB
ResNet-50 frozen	1×10^{-3}	15	5	ImageNet
ResNet-50 fine-tuned	1×10^{-4}	20	5	ImageNet

Table 4: Training settings. Weight decay is 10^{-4} for all models.

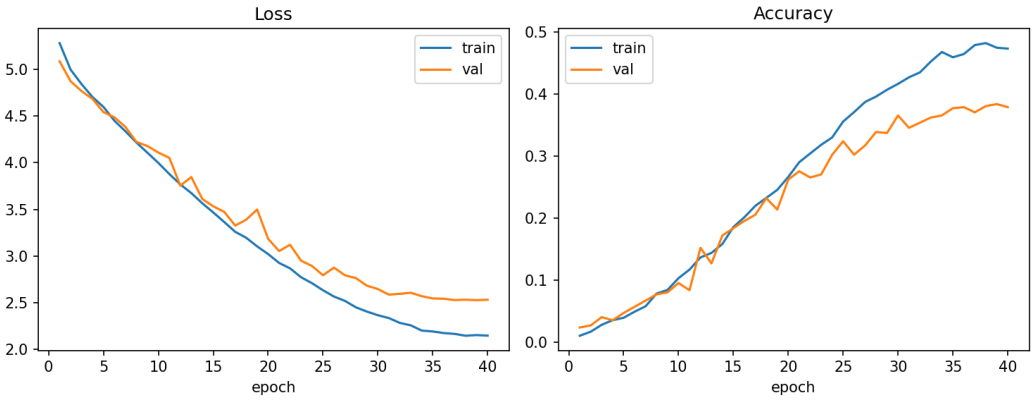
5 Results

5.1 Scratch CNN vs transfer learning

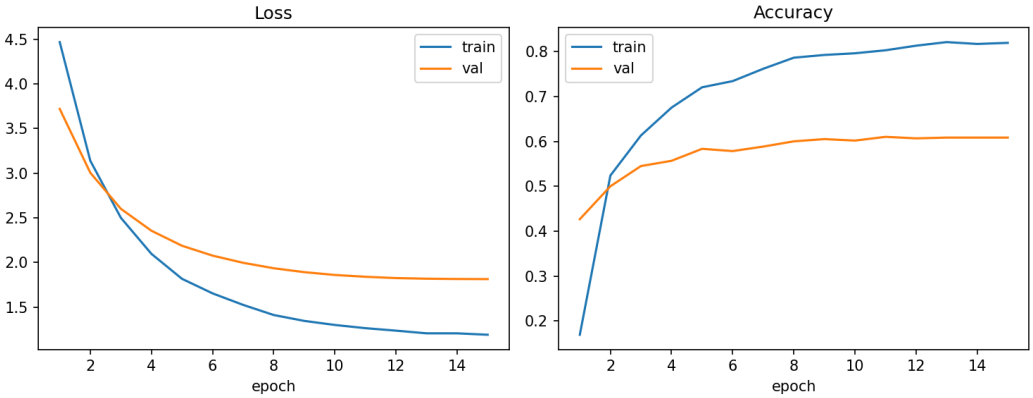
Table 5 summarizes the results on the test set and Figure 7 shows the training curves.

Model	Trainable	Train acc	Best val acc	Test acc	Macro F1	Time
Scratch CNN	4.82M	47.3%	38.3% (ep. 39)	40.2%	38.8%	–
ResNet-50 frozen	0.41M	81.9%	61.0% (ep. 11)	64.3%	64.0%	9.6 min
ResNet-50 fine-tuned	23.9M	99.1%	**82.0%** (ep. 16)	**82.9%**	**82.9%**	13.3 min
ResNet-50 tuned	23.9M	99.4%	81.3% (ep. 11)	81.8%	81.8%	10.7 min

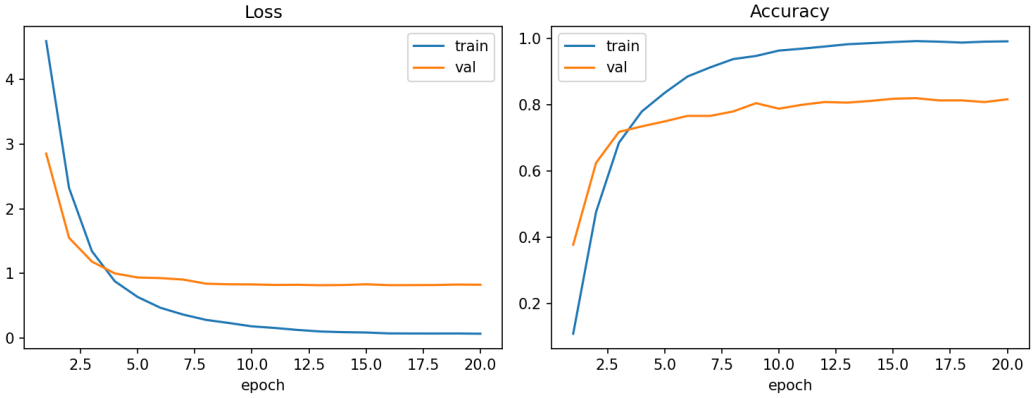
Table 5: Results on the test set (5,794 images). Random guessing would give 0.5%.



(a) Scratch CNN



(b) ResNet-50, frozen backbone



(c) ResNet-50, fine-tuned

Figure 7: Training and validation loss and accuracy for the three models.

Scratch CNN (40.2%). This is about 80 times better than chance, but still low. With only ∼27 images per species and no prior knowledge, the network cannot really learn the small details that separate similar species. The gap between train (47.3%) and validation (38.3%) is only about 9 points, so the model is underfitting more than overfitting. Early stopping never triggered because the validation accuracy was still slowly going up at epoch 40, but by then the cosine schedule had brought the learning rate to almost zero. The top-5 accuracy was 72%, so the right species is usually among the first guesses.

Frozen ResNet-50 (64.3%). Just by training one linear layer on top of ImageNet features we gained 24 points, which shows how much the pretrained features already help. The validation

accuracy stopped improving after epoch 11, because a single linear layer on fixed features does not have enough capacity to go further.

Fine-tuned ResNet-50 (82.9%). Letting all the layers adapt to birds added another 18.6 points and more than doubled the scratch CNN. The training accuracy reached 99% while validation stayed around 82%, so the model basically memorized the training set. However, the validation loss did not go up, so this did not hurt generalization. Macro F1 is equal to accuracy, which means the model is not only doing well on species that have more test images.

5.2 Hyperparameter tuning

We tuned the learning rate and weight decay of the fine-tuned ResNet-50 with both grid search and random search. Each trial ran for 5 epochs with the same seed and used only the train and validation sets. The grid was lr $\in \{10^{-4}, 3 \times 10^{-4}\} \times$ wd $\in \{10^{-4}, 10^{-3}\}$. For random search we sampled 4 points log-uniformly between the same limits, since sampling from the same 4 grid points would just be the grid in a different order [7].

Method	Best lr	Best wd	Best val acc (5 ep.)	Trials	Time
Grid	3×10^{-4}	1×10^{-4}	79.5%	4	12.5 min
Random	1.35×10^{-4}	1.67×10^{-4}	79.7%	4	12.6 min

Table 6: Grid search vs random search.

We then trained the best setting (from random search) for the full schedule. It got 81.3% validation and 81.8% test accuracy, which is actually *a bit worse* than our default setting (82.0% / 82.9%). We think there are three reasons for this:

1. The trials were only 5 epochs long, and the best setting after 5 epochs is not necessarily the best after 20.
2. All the trials with lr between 1.35×10^{-4} and 3×10^{-4} ended between 78.2% and 79.7%. On 600 validation images, 1 point is only 6 images, so these differences are basically noise.
3. Our default lr of 10^{-4} was already in the good range.

Since we select models by validation accuracy and not test accuracy, we kept the untuned fine-tuned ResNet-50 as our final model. With only 2 hyperparameters and 4 trials, grid and random search performed the same; we expect random search to be more useful when there are more hyperparameters.

6 Error Analysis and Grad-CAM

6.1 Most confused species

Using the confusion matrix on the test set, we listed the 5 species pairs that each model mixes up the most, counting mistakes in both directions (Table 7).

#	Scratch CNN	ResNet-50 fine-tuned
1	Eared Grebe ↔ Horned Grebe (13)	Long-tailed Jaeger ↔ Pomarine Jaeger (15)
2	Arctic Tern ↔ Common Tern (13)	Chuck-will's-widow → Whip-poor-will (14)
3	Loggerhead Shrike ↔ Great Grey Shrike (12)	Brandt's Cormorant ↔ Pelagic Cormorant (14)
4	Arctic Tern ↔ Forster's Tern (12)	Acadian Flycatcher ↔ Yellow-bellied Flycatcher (12)
5	Blue-winged Warbler ↔ Prothonotary Warbler (12)	Arctic Tern ↔ Common Tern (12)

Table 7: The 5 most confused species pairs on the test set (number of mistakes).

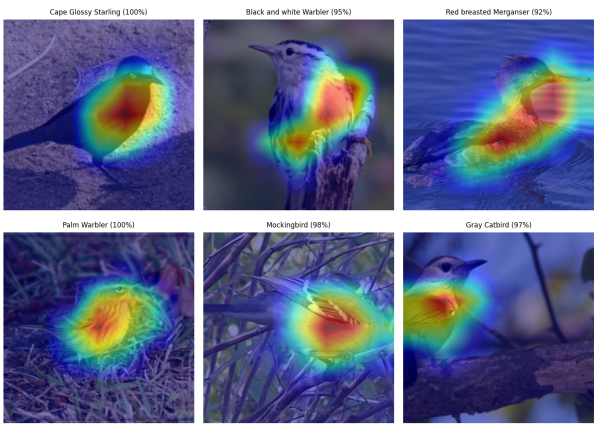
All of the ResNet-50 pairs are species from the same family that look almost identical, and that even bird watchers often separate by their song, so we consider these "reasonable" mistakes. Interestingly, the Chuck-will's-widow mistake only goes one way: the model labels both species as Whip-poor-will. The scratch CNN, on the other hand, also confuses species that only share a color, like the Blue-winged and Prothonotary Warblers, which are both yellow but otherwise look different. This suggests it relies on color more than on shape.

6.2 Grad-CAM

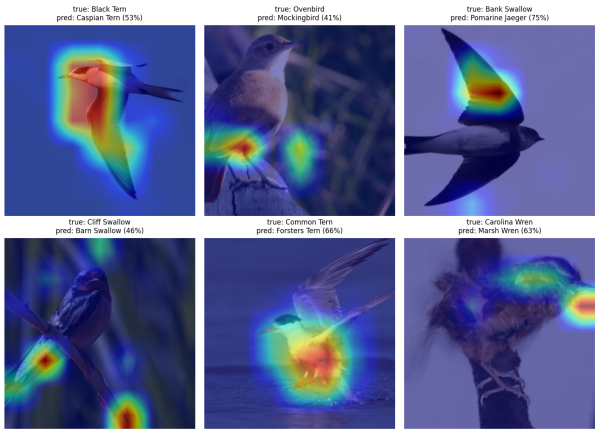
We implemented Grad-CAM [4] ourselves using PyTorch hooks. A forward hook on the last convolutional layer stores the feature maps A^k, and a hook on the output tensor stores their gradients when the class score y^c is backpropagated. The weight of each channel is the average of its gradient, and the heatmap is

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \qquad L^c_{\text{Grad-CAM}} = \text{ReLU}\Big(\sum_k \alpha_k^c A^k \Big), \tag{1}$$

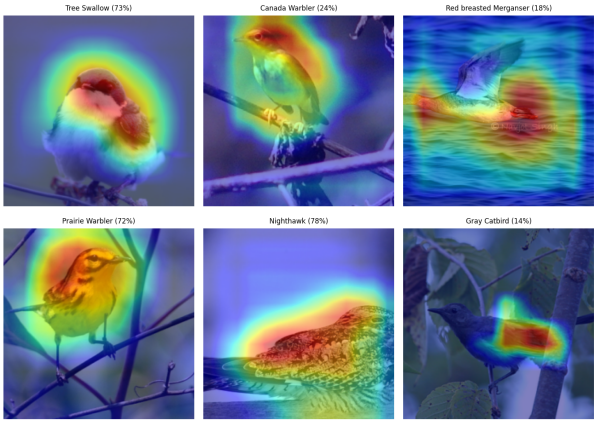
which we then resize to 224×224 and normalize to $[0, 1]$. One problem we ran into is that for the frozen ResNet-50 no gradient flows to `layer4` because none of its weights need gradients, so the hook never fired. We fixed this by setting `requires_grad` on the input image.



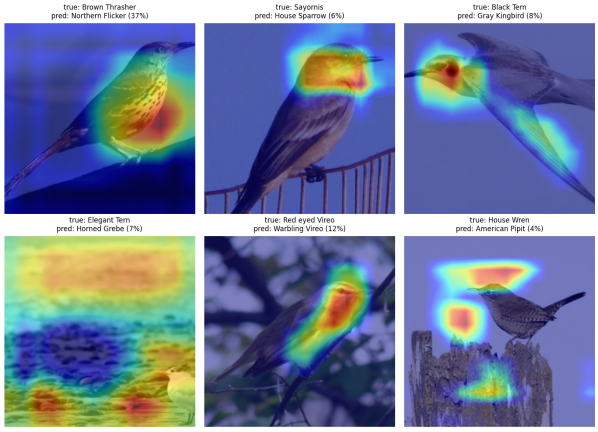
(a) ResNet-50, correct predictions



(b) ResNet-50, wrong predictions



(c) Scratch CNN, correct predictions



(d) Scratch CNN, wrong predictions

Figure 8: Grad-CAM heatmaps on random correct and wrong test predictions.

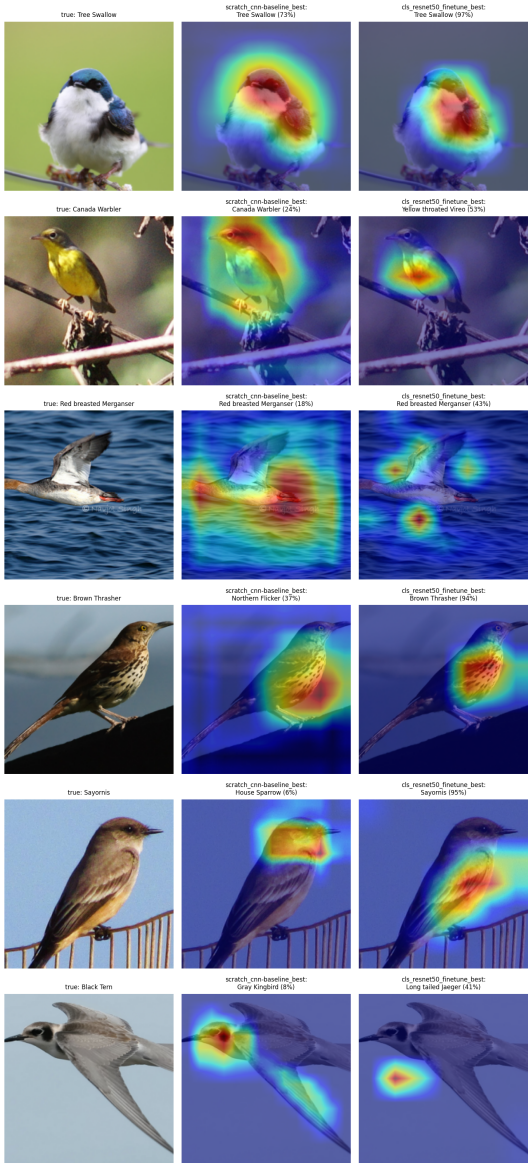


Figure 9: The same test images explained by both models. Left: input, middle: scratch CNN, right: fine-tuned ResNet-50.

From Figures 8 and 9 we noticed the following:

- **ResNet-50, correct (92–100% confidence):** the heatmap is on the body, breast and wing pattern of the bird, for example the stripes of the Black-and-white Warbler. The background stays cold.
- **ResNet-50, wrong (41–75% confidence):** the confidence is clearly lower, so the model is "aware" that it is unsure. Most mistakes are lookalikes (Common Tern → Forster's Tern, Cliff Swallow → Barn Swallow). In a few cases the attention is off the bird, such as on the legs and perch of an Ovenbird. One Bank Swallow photographed from below in flight was called a Pomarine Jaeger, and the heatmap shows it was looking at the wing silhouette, which is the only thing visible against the sky.
- **Scratch CNN, wrong (4–37% confidence):** it often looks at the *background*. On an Elegant Tern the heat covers the water and the model predicts Horned Grebe, which is also a water bird. On a House Wren it looks at the sky. So with little data and no pretraining it learned shortcuts like "water means water bird".
- **Side by side:** on images where the scratch CNN fails (e.g. Brown Thrasher), ResNet-50 is

right with about 95% confidence and focuses on the spotted breast. However, Grad-CAM is not a proof that the model is right for the right reason: in one case ResNet-50 got a Red-breasted Merganser right with its attention scattered over the water.

7 Comparison with Published Results

We compared our final model with published ResNet-50 results on the same official test split, using the same metric (top-1 accuracy) and the same kind of supervision (image labels only, no bounding boxes or part annotations).

Model	Input	Epochs	Test acc
Ours, ResNet-50 fine-tuned	224	20	**82.9%**
ResNet-50, Li et al. [8]	224	90	81.5%
ResNet-50, Li et al. [8] / NTS-Net baseline [9]	448	90	84.5%
ResNet-50, Xie et al. [10]	448	95	85.3%
NTS-Net, Yang et al. [9]	448	–	87.5%

Table 8: Our final model compared with published results on CUB-200-2011.

At the same input size of 224 px, our result matches the published ResNet-50 (81.5%). Our small lead can be explained by the stronger ImageNet weights we use (`IMAGENET1K_V2`, 80.9% on ImageNet compared to about 76% for the original weights), and also by noise, since the 95% confidence interval on our test accuracy is about ±1 point. The papers that report 84.5–85.3% use 448 px images, and Li et al. showed that going from 224 to 448 alone adds 3 points. This makes sense because the details that separate species only cover a few pixels at 224. We also train on 10% fewer images (because of our validation set) and for fewer epochs, and we do not use any part-localization tricks like NTS-Net.

8 Limitations and Problems We Faced

- **Tuning did not help.** The 5-epoch trials were too short to rank the settings correctly, and the validation set is too small to tell apart settings within 1–2 points. Next time we would use longer trials or repeat each trial with different seeds.
- **The scratch CNN plateaued at 40%** and uses background shortcuts. Stronger augmentation (e.g. CutMix), longer training or residual connections could help.
- **224 px input limits the fine details.** Training at 448 px is the main thing we would try next, although it costs about 4 times more compute.
- **Grad-CAM is coarse.** The maps are only 7×7 before resizing, so they show regions and not small parts like the eye ring.
- **No "unknown" option.** The classifier always picks one of the 200 species, even for a bird that is not in CUB.

We also ran into some engineering problems along the way. MLflow's default database path broke on Windows when the user folder name had a space in it, so we set the tracking URI ourselves. The `data/` rule in `.gitignore` accidentally hid `src/data/` as well. Two ResNet-50 runs had the same name and overwrote each other's checkpoint, so we started naming runs after their config file. Finally, we found out that the species list in the dataset is not in alphabetical order, so we had to save the class names in label order and never sort them.

9 Conclusion

In this milestone we built a full pipeline for bird species classification on CUB-200-2011, from data analysis to explainability. The EDA helped us catch real issues such as train–test duplicates and noisy captions before training. Our results clearly show the value of transfer learning on a small fine-grained dataset: the accuracy went from 40.2% for the scratch CNN to 64.3% with a frozen ResNet-50 and 82.9% after fine-tuning, which is in line with published results at the same resolution. Grad-CAM confirmed that the pretrained model looks at the bird while the scratch CNN often looks at the background. Our final model is the fine-tuned ResNet-50, and its encoder will be the backbone for the colorization and captioning tasks in Milestone 2.

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